

AI Content Generator Generator Cookbook

M.Sc. Applied Data Science & Artificial Intelligence

SRH Hochschule Heidelberg

Namrata, Nihal, Pramod, Anuj, Gourav

Contents

1. Introduction
2. The System
3. Choosing the Models
 - 3.1 Providers Evaluated
 - 3.2 Metrics Per Modality
 - 3.3 Why Cloudflare for the Live System
 - 3.4 Reliability Issues We Found
 - 3.5 Free-Tier Friction
 - 3.6 Phase-1 Results
 - 3.7 Software Problems Fixed During Scoring
 - 3.8 Correcting the Audio Measurement
4. Per-Modality Quality Scores
 - 4.1 Fixed External Goalposts
 - 4.2 The quality_pair Formula
 - 4.3 The CLAP Investigation
 - 4.4 The λ Sweep
 - 4.5 Phase-2 Results
5. LLM Judge and Learned Weights
 - 5.1 The LLM Judge
 - 5.2 Learned Weights (Linear Regression)
 - 5.3 Quality-Weighted Coherence (WTSAS)
6. Cross-Modal Coherence
 - 6.1 The Gap Every Earlier Metric Leaves
 - 6.2 The Starting Idea
 - 6.3 Version 1 — Everything Through ImageBind
 - 6.4 Version 2 — A Mixed Set of Specialist Models
 - 6.5 What the Numbers Said
 - 6.6 The Professor's Feedback
7. New Scoring Methods Under Test
 - 7.1 Metric-to-Grade Normalisation
 - 7.2 Quality-Weighted TSAS (WTSAS)
 - 7.3 Naive Bayes Posterior
 - 7.4 Likelihood-Ratio Score
 - 7.5 Mismatched-Pair Baseline
 - 7.6 Text-Mediated Coherence Check
8. Comparing the Methods — Real Results
 - 8.1 Raw Per-Feature Signal
 - 8.2 Final Comparison
 - 8.3 Why Naive Bayes Cannot Beat WTSAS
9. Related Work
10. Which Paper We Adopted What From
11. Issues Found and Fixed — Running Log
12. What Is Still Missing and Next Steps
 - 12.1 Methodology
 - 12.2 Concrete To-Do Before the Next Review

1. Introduction

This cookbook is the working record of how we built a system that turns one short prompt into three different kinds of content at once — a written passage, an image, and a spoken narration of that passage — and, just as importantly, how we learned to measure whether those three outputs actually belong together. It is written the way a good recipe book is written: not as a polished afterthought, but as an honest account of what we tried, what broke, what we fixed, and what the numbers finally told us.

Each output is produced independently by its own model, which is what we mean by a multi-agent design. The text comes from one model, the image from another, the audio from a third, and only at the very end are the three shown together as a single result. A video modality is planned for later and is not part of this build.

The project came together in three movements. First we built the generation pipeline and the application around it — the machinery that takes a prompt, calls the right models, and returns the combined output. Then we settled which provider should sit behind each modality, which took real benchmarking rather than guesswork. Finally, and this is where most of the recent effort has gone, we built a way to evaluate the result: not just “is the text good” or “is the image good” on their own, but the harder question of whether the text, the image, and the audio all describe the same thing.

That last question turned out to be surprisingly deep, and chasing it properly is what separates this edition from the first. The sections that follow move from the system itself, through model selection and the practical realities of running on a free tier, into per-modality quality scoring, an LLM-judge calibration loop, cross-modal coherence, and finally the family of scoring methods we tested before settling on one to bring to the professor.

2. The System

The central deliverable is a working content generator. A user opens the Streamlit application, enters a prompt such as “a cozy wooden cabin beside a calm mountain lake at sunrise,” and receives back three things: a descriptive paragraph, a generated image, and a spoken version of the paragraph. Each of these comes from a separate model, and the three are produced from the same input and then combined at the end.

The models that power the live system are shown below. They were chosen after the benchmarking described in Section 3, and they all sit behind a single Cloudflare account.

Modality	Model in the live system	Cloudflare endpoint
Text	Llama 3.1 8B Instruct	@cf/meta/llama-3.1-8b-instruct
Image	FLUX.1 [schnell]	@cf/black-forest-labs/flux-1-schnell
Audio	MeloTTS	@cf/myshell-ai/melotts

Everything else in this cookbook exists to make that one application produce better content, more cheaply, and more reliably. The provider comparisons, the scoring pipeline, and the long list of fixes were all in service of that single goal.

3. Choosing the Models

Before committing to anything, we compared providers across all three modalities. Two genuinely different philosophies were on the table, and each came with an honest trade-off.

The first approach uses a separate, specialised provider for each modality. This buys you best-in-class models in every category, but every provider then needs its own account, its own API key, and its own billing. Powering all three modalities this way would have meant building and maintaining more than ten separate integrations. The second approach uses a single provider for everything: Cloudflare Workers AI hosts text, image, and audio models behind one account and one key.

3.1 Providers Evaluated

We ran the same 56 prompts through every candidate and scored the outputs with a fixed metric set. The large commercial models — Claude, ChatGPT, and Gemini — were kept only as a reference column, using figures published in 2025 papers rather than re-benchmarked by us, because the point was to place our system in context, not to re-measure the giants.

Modality	Providers considered
Text	Groq (Llama 3.3 70B), Mistral (Mistral Large), Cerebras (GPT-OSS 120B), Cloudflare (Llama 3.1 8B)
Image	Black Forest Labs (FLUX Pro), Stability AI (SDXL), Hugging Face (SD 1.5), Pixazo, FLUX.1-schnell, Leonardo Lucid Origin
Audio	Cloudflare/Deepgram Aura, Camb AI, Cartesia, Fish Audio, MeloTTS, Gemini TTS

3.2 Metrics Per Modality

Modality	Metrics used
Text	BERTScore, ROUGE-L, Readability (Flesch–Kincaid grade / 10)
Image	CLIP similarity, CLIPScore, Aesthetic (LAION predictor, 0–10)
Audio	CLAP, MOS (DNSMOS, 1–5), Word Error Rate (Whisper)

3.3 Why Cloudflare for the Live System

We chose Cloudflare Workers AI, and the reasons were practical rather than ideological.

- **One key.** One key covers all three modalities, which is the single biggest simplification in the whole project.
- **Genuinely free to build on.** The free tier gives 10,000 “neurons” per day — Cloudflare’s own unit of compute — which was enough to build and test everything without ever setting up billing. A full 56-prompt run costs roughly 4,350 neurons, about two runs a day.
- **Simple and fast.** There is one integration to maintain instead of one per provider, and because inference runs on Cloudflare’s edge network, each generation comes back in under two seconds.
- **The catch.** The daily quota is a real ceiling, and we hit it more than once. For any further work at this volume, a funded paid API is the honest recommendation.

3.4 Reliability Issues We Found

Three undocumented platform behaviours surfaced during testing, each of which would have quietly caused trouble later.

- **MeloTTS is flaky.** Roughly a quarter of MeloTTS calls return an HTTP 500. We confirmed it was Cloudflare’s side, not ours, by sending the same request eight times — two failed, with no pattern tied to the input. A three-try retry with backoff brought audio up to effectively full reliability, and the same retry now lives in the app so a live generation never fails on the audio step.
- **Images are secretly JPEGs.** The image endpoint returns JPEG bytes in files named .png. Harmless, because libraries detect the real format, but worth knowing for any code that assumes PNG.
- **44.1 kHz, not 48.** The generated audio is sampled at 44.1 kHz, not the 48 kHz we had assumed, which matters for the scoring step because each scorer expects its own rate.

3.5 Free-Tier Friction

- **Quota.** The 10,000-neuron daily limit was reached more than once — a full run plus a few scoring experiments empties it in an afternoon.
- **A leaked key.** An API key briefly leaked and Cloudflare’s own leaked-credential detection auto-disabled it. An unwelcome but useful reminder never to commit keys; the key was regenerated.
- **The dashboard lied.** The dashboard reported zero usage while the API kept returning quota errors for about fourteen hours past the documented midnight reset. We never fully explained it, and it cost us time.

3.6 Phase-1 Results (averages over 56 prompts)

Measure	Ours	Claude	ChatGPT	Gemini
Text — BERTScore	0.844	0.851	0.832	0.796
Text — ROUGE-L	0.056	0.078	0.071	0.068
Text — Readability	0.948	1.85	1.72	1.68
Image — CLIP	0.31	0.312	0.305	0.298
Image — Aesthetic (/10)	5.80	7.20	7.05	6.90
Audio — CLAP	0.082	0.74	0.72	0.76
Audio — MOS (/5)	3.37	4.10	4.05	4.20
Audio — WER (corrected)	0.095	0.072	0.085	0.068

Three further figures simply are not published for the commercial models, and they matter: our system responds in under two seconds per modality, a full run stays inside the free daily quota, and the image question-answering check confirmed about 90 percent factual accuracy.

3.7 Software Problems Fixed During Scoring

- **A crashing image scorer.** The image scorer crashed on every image because it was written for an older transformers version whose output shape had since changed — every image scored zero until we made the code work with both versions.

3.8 Correcting the Audio Measurement

The word error rate started at 1.0 for every clip, which would imply the speech was gibberish. The real cause was mundane: the transcript was being compared to the short original prompt, when the audio actually reads out the longer generated paragraph. Comparing two different texts guarantees the worst possible score. Once the comparison used the text that was actually spoken, average WER fell to about 0.095 — roughly 90 percent of words transcribed correctly — turning a meaningless number into a real measure of intelligibility.

4. Per-Modality Quality Scores

The Phase-1 metrics live on different scales — BERTScore runs 0 to 1, aesthetics 0 to 10, MOS 1 to 5, and WER is lower-is-better. Before they can be compared or combined they have to be brought onto one footing. Phase 2 converts everything into a single 0-to-1 quality grade per modality.

4.1 Fixed External Goalposts

Each raw metric is mapped to 0–1 against a fixed external standard, deliberately not against our own data spread, so the grades mean the same thing run to run.

Metric	Goalpost
MOS	ITU 1–5 speech-quality scale
Aesthetic	LAION 1–10 scale
WER / BERTScore	already 0–1
CLIP	floor = mismatched-pair baseline (Section 8)

4.2 The quality_pair Formula

The heart of per-modality scoring is a small formula that combines two related sub-scores while punishing imbalance between them:

$$\text{quality_pair}(a, b) = \text{avg}(a, b) - \lambda \times \text{variance}(a, b), \quad \lambda = 0.5$$

The average rewards both sub-scores being high; the variance penalty marks down a pair where one channel is strong and the other weak, on the principle that one good channel should not paper over a bad one. The three modality grades are built from it as follows:

$Q_{\text{text}} = \text{BERTScore}(\text{generated_text}, \text{prompt})$

$Q_{\text{image}} = \text{quality_pair}(\text{CLIP_score}, \text{aesthetic_score})$

$Q_{\text{audio}} = \text{quality_pair}(\text{semantic_score}, 1 - \text{WER})$

Here semantic_score is BERTScore between the Whisper transcript of the audio and the prompt. The reason it replaces the more obvious CLAP score is worth its own section.

4.3 The CLAP Investigation — Why CLAP Was Dropped

CLAP was the natural audio-versus-prompt metric, the audio analogue of CLIP. In practice it failed badly on our data: it returned exactly 0.0 for roughly half the prompts, because it was trained on environmental sounds and music, not speech, and our audio is text-to-speech narration. It was listening for sound events and hearing none. We replaced it with a two-step semantic approach — transcribe the audio with Whisper, then BERTScore that transcript against the prompt — which gives usable signal on every single prompt. CLAP is kept only as a comparison column, to document honestly why it was not suitable. With the change, average audio quality rose from near-zero to a meaningful 0.464.

4.4 The λ Sweep

We swept λ from 0.00 to 2.00 in steps of 0.05, scoring each value by the Pearson correlation between the recomputed quality scores and the LLM-judge overall score. The correlation stays weak throughout and the curve is nearly flat, which tells us plainly that λ is a minor knob — no value turns the per-modality quality score into a strong predictor of judged quality on its own.

λ	Pearson r
0.00	0.375
0.50 (our default)	0.399
1.00	0.422
1.80 (peak)	0.442
2.00	0.441

4.5 Phase-2 Results (means over 53 prompts)

Quantity	Mean	Notes
Q_text	0.848	consistently high, low spread
CLIP (raw)	0.346	
Aesthetic (0–1)	0.699	
Q_image	0.504	
semantic_score	0.855	
1 – WER	0.186	suspiciously low — see Section 11
Q_audio	0.464	dragged down by the WER term

5. LLM Judge and Learned Weights

Weighting the three modality grades equally was a guess, and guesses are worth replacing with evidence. Phase 3 does that by learning the weights from an LLM judge that stands in for a human evaluator.

5.1 The LLM Judge

Gemini is shown the prompt, the generated text, the generated image, and the audio transcript, and returns an overall score plus per-modality scores from 1 to 10 with a one-sentence reason. All 53 prompts were rated. It rates text and image highly and audio much lower — reassuringly, the same shape as our own per-modality scores.

Judge score	Mean	Min	Max
Overall	6.81	4	8
Text	8.11		
Image	8.25		
Audio	4.66		

Getting the judge working took two rounds of trial and error worth recording: the first attempt imported the old `google.generativeai` package instead of the newer `google.genai` and failed outright, and the second cut its own JSON response off mid-sentence because the token limit was set too low. Raising the limit fixed it.

5.2 Learned Weights (Linear Regression)

With the judge’s overall score as the target and the three quality grades as features, a linear regression learns how much each modality actually matters to perceived quality.

Weight	Value (normalised)
w1 — Q_text	0.287
w2 — Q_image	0.520
w3 — Q_audio	0.193

Fit	Value
R ² (train)	0.159
RMSE (train)	0.93
R ² (leave-one-out CV)	0.04
RMSE (LOO CV)	0.99

The honest read is that image quality carries the most weight, but the model barely generalises — a leave-one-out R² of about 0.04 is close to no predictive power at all. With only 53 samples and three features that mostly cluster near their means, these weights are an indicative signal, not a finished result. More prompts and more runs per prompt are needed before they can be trusted, and we say so plainly rather than dressing it up.

5.3 Quality-Weighted Coherence (WTSAS)

The learned weights are not an end in themselves; they feed into a quality-weighted version of the coherence score. Each pairwise similarity is scaled by the average learned quality of the two modalities it connects, so a similarity between two high-quality outputs counts for more than one between a strong and a weak output.

$$s1_weighted = s1 \cdot (w1 \cdot Q_text + w2 \cdot Q_image) / (w1 + w2)$$

$$s2_weighted = s2 \cdot (w1 \cdot Q_text + w3 \cdot Q_audio) / (w1 + w3)$$

$$s3_weighted = s3 \cdot (w2 \cdot Q_image + w3 \cdot Q_audio) / (w2 + w3)$$

$$\text{WTSAS} = (s1_weighted + s2_weighted + s3_weighted) / 3$$

$$\text{Final} = \text{WTSAS} - \lambda \cdot \text{variance}(s1_weighted, s2_weighted, s3_weighted)$$

6. Cross-Modal Coherence

6.1 The Gap Every Earlier Metric Leaves

BERTScore only ever looks at the text. CLIP only looks at the image against the text. CLAP only looks at the audio against the text. Not one of them asks whether the three outputs agree with each other. A generator could score beautifully on all three per-modality metrics and still hand back a mismatched set — a librarian owl in the text, a plain barn owl in the image, and narration about something else entirely — precisely because the three generators never see each other’s work. To catch that, we need a score computed between the three outputs, not against the prompt.

6.2 The Starting Idea

Encode all three outputs into one shared vector space, take the cosine similarity of each pair, average them into a tri-modal alignment score, and penalise disagreement between the three:

$$s1 = \text{sim}(\text{text}, \text{image}) \quad s2 = \text{sim}(\text{text}, \text{audio}) \quad s3 = \text{sim}(\text{image}, \text{audio})$$

$$\text{TSAS} = (s1 + s2 + s3) / 3$$

$$\text{Coherence} = \text{TSAS} - \lambda \cdot \text{variance}(s1, s2, s3)$$

If all three agree, the similarities sit close together, the variance is near zero, and the score is simply TSAS. If one modality drifts, its two pairs fall, the variance rises, and the score is pulled below TSAS. λ sets how harshly disagreement is punished.

6.3 Version 1 — Everything Through ImageBind

Field	Detail
Script	<code>imagebind_score.py</code>
Model	ImageBind-huge, pretrained (1024-dim shared space)
Modalities	text, image, audio — all in the one space
λ tested	0.1, 0.5, 1.0, 2.0
Prompts	56 attempted, 53 scored (13, 17, 56 skipped — no image)
Output	<code>imagebind_coherence_scores.csv</code>

Audio is the fiddly part. The waveform is resampled to 16 kHz, turned into a 128-bin log-mel spectrogram, and padded or cropped to 204 frames before ImageBind’s audio encoder — which was trained to treat spectrograms as images — will accept it. Text and image go through their own ImageBind encoders directly. All three come out as 1024-dimensional vectors, and that shared dimensionality is exactly what makes a single cosine-similarity function meaningful across every pair.

6.4 Version 2 — A Mixed Set of Specialist Models

Field	Detail
Script	<code>multimedia_score.py</code>
Text ↔ image	CLIP
Text ↔ audio	CLAP
Image ↔ audio	Whisper transcribes the audio, then CLIP compares transcript ↔ image
λ	0.5 fixed
Prompts	first 5 only (proof of concept)

The image-to-audio pair is the interesting one. Instead of comparing a CLIP embedding and a CLAP embedding that were never trained against each other, it transcribes the audio into text and does a CLIP text-to-image comparison it was actually trained for. That sidesteps the awkward question of whether CLIP’s and CLAP’s spaces even mean the same thing — at the cost of depending on Whisper’s transcription being accurate.

6.5 What the Numbers Said (ImageBind, 53 prompts)

Similarity	Mean
s1 — text ↔ image	0.362
s2 — text ↔ audio	0.106
s3 — image ↔ audio	0.116
TSAS	0.195
variance	0.016

The text-to-image similarity averages roughly three times higher than either audio pair, and it does so on every one of the 53 prompts — not a handful of outliers. Two prompts even produced a slightly negative similarity, with nothing obviously broken in either generation. The gap is far too uniform to be about content; it shows up whether the prompt is a librarian owl or a taco lifting weights.

It looks structural. ImageBind’s audio encoder simply sits further from the text and image encoders in the shared space than text and image sit from each other, independent of whether the audio actually matches. And because the coherence score is computed on these three numbers directly, and s1 is structurally largest for every prompt, the variance penalty is partly measuring a fixed quirk of the embedding model rather than how well a given triple agrees. A perfectly matched triple gets penalised for the same reason a mismatched one does. Raising λ makes this worse, not better: at $\lambda = 2.0$ the average score drops to about half of average TSAS, purely from a largely structural penalty.

6.6 The Professor’s Feedback

The professor’s verdict was direct: this TSAS-minus-variance method is a reasonable first cut but basic, because it leans on raw cosine similarities that are not comparable across different encoders. The instruction was to try several better-grounded methods, measure each against a human or LLM sense of whether the outputs match, and bring the best one back for approval. That is exactly what the next two sections do.

7. New Scoring Methods Under Test

These methods are implemented or specified and are being evaluated on the same 53 triples. The plan is to compare each against the LLM-judge overall score and a small hand-labelled batch, then adopt the one that tracks human judgement best.

7.1 Metric-to-Grade Normalisation

Already in use. Every raw metric is mapped to 0–1 against an external standard, then combined within and across modalities by a geometric mean so that a weak modality genuinely drags the whole score down:

Text = BERTScore

Image = $\sqrt{(\text{CLIP} \times \text{Aesthetic})}$

Audio = $\sqrt{(\text{MOS} \times \text{WER_grade})}$

Final = $(\text{Text} \times \text{Image} \times \text{Audio})^{(1/3)}$

7.2 Quality-Weighted TSAS (WTSAS)

Scales each pairwise similarity by the learned quality of the two modalities it connects, then applies the variance penalty, using the Phase-3 regression weights (0.29 text, 0.52 image, 0.19 audio). This is our main novel metric.

7.3 Naive Bayes Posterior

Replaces the hand-tuned score with a probability. The LLM-judge score is binned into Good, Medium, and Bad; a Gaussian Naive Bayes is fitted on the three quality features; and the continuous posterior probability of “good” — not the hard label — becomes the new score, validated with leave-one-out cross-validation.

7.4 Likelihood-Ratio Score

Taken from biometric score-fusion theory. For each feature, fit one Gaussian to matched triples and one to mismatched (shuffled) triples, and sum the log-likelihood-ratios. Summing log-LR over independent features is exactly Naive Bayes, so 7.3 and 7.4 are two views of the same underlying mathematics — one calibrated on a good/bad label, the other on a matched/mismatched contrast.

7.5 Mismatched-Pair Baseline (the “CLIP Floor”)

Shuffle prompts against outputs to build a distribution of similarity scores for triples that are known not to match. Every raw similarity is then read relative to this floor rather than on its own, which is what subtracts out the structural modality gap identified in Section 6.5.

7.6 Text-Mediated Coherence Check

Caption the generated image, transcribe the generated audio, and have an LLM check whether the prompt’s key attributes actually appear in both descriptions. Unlike the embedding score, this one names what is missing — “glasses absent from the image” — rather than only reporting that a number went down.

8. Comparing the Methods — Real Results

This is the part that matters most, because it is where the methods stop being ideas and start being numbers. Every candidate was run on the real repository data for all 53 prompts, and each was correlated against the LLM-judge overall score. We report Spearman and Kendall correlations rather than Pearson, because at this sample size a rank-based measure is far more stable than one sensitive to outliers.

8.1 Raw Per-Feature Signal

Before any model, it is worth asking which features carry signal at all. Two of the three do; the third does not.

Feature	Spearman	p-value	Verdict
q_text	+0.316	0.021	usable
q_image	+0.297	0.031	usable
q_audio	−0.006	0.969	no signal

The audio feature is essentially uncorrelated with judged quality. This is not a small detail — it explains a great deal of what follows, because any method that is forced to weight all three features equally will be dragged down by a feature that predicts nothing.

8.2 Final Comparison (n = 53)

Method	Spearman	Kendall	Pearson
WTSAS	0.451	0.350	0.399
Likelihood Ratio	0.312	0.240	0.259
Naive Bayes	0.296	0.231	0.241
TSAS (plain averaging)	−0.077	−0.056	−0.085

WTSAS wins clearly at this sample size. The two probabilistic methods, Likelihood Ratio and Naive Bayes, land close to each other — which is exactly what their shared mathematics predicts — and both were held back by the same dead audio feature and the same small-class instability, not by any flaw in the probabilistic framing itself. Plain TSAS comes out negative, which confirms the original critique: averaging incommensurable cosine similarities tracks nothing useful.

8.3 Why Naive Bayes Cannot Beat WTSAS

The professor asked directly why the probabilistic methods underperform, and there are three concrete reasons rather than one vague one.

- **Binning throws away what we are graded on.** Naive Bayes needs the continuous judge score chopped into Good/Medium/Bad classes before it can fit anything, but we evaluate with Spearman, which rewards getting the fine-grained order right. A 7.1 and a 7.9 can fall in the same bin, and Naive Bayes can no longer tell them apart. WTSAS fits directly on the raw continuous score and never loses that ordering.
- **The smallest class has four examples.** Quantile binning leaves the smallest class with just four examples, and a Gaussian fitted from four points is unstable rather than informative. The Naive Bayes correlation swung between 0.05 and 0.30 simply from changing which features were included — a sign of noise, not structure.
- **One feature carries no signal.** Naive Bayes multiplies per-feature likelihoods and has no way to down-weight a dead feature, so including q_audio actively pulled the score down every time.

WTSAS, being a regression, simply learns a low weight for a feature that does not predict the target.

9. Related Work

To place the project properly, we surveyed a few neighbouring areas. On the academic side, the closest systems are MM-StoryAgent, which builds narrated storybook video from cooperating agents; MAGUS, which coordinates several models without retraining any of them; and NExT-GPT, an any-to-any model. The metrics we rely on trace back to well-established papers, including the original CLIPScore work, the LAION aesthetic predictor, and EmergentTTS-Eval for speech.

On the open-source side, projects such as MoneyPrinterTurbo and ShortGPT are reassuring evidence that the general architecture is well founded. Commercially, Synthesia, Pictory, Lumen5, and NotebookLM each cover part of the same ground, while Artificial Analysis benchmarks models but treats every modality separately. The specific gap we sit in is narrow but real: building multimodal generators is not new and the individual metrics are standard, but building a working system on a single free edge provider and then measuring its output carefully across all three modalities at once — including cost, speed, and reliability, and going further to score whether the three outputs are mutually coherent — is uncommon.

10. Which Paper We Adopted What From

The literature review is easiest to read as a single map: each paper, what it does in plain terms, and the concrete thing we carried into our own work.

#	Paper	What it does	What we took
1	ImageBind (Girdhar et al., FAIR/Meta, CVPR 2023)	Learns one joint embedding across six modalities; only image-paired data is needed to bind them.	The entire Version-1 coherence scorer — text, image, and audio in one 1024-D space — plus the audio-as-spectrogram preprocessing.
2	An Image is Worth 16×16 Words / ViT (Dosovitskiy et al., ICLR 2021)	A pure transformer on 16×16 image patches beats CNNs at scale.	The image → vector explanation (224×224 → 196 patches → 768 → 1024-D); the encoder mechanism inside both CLIP and ImageBind.
3	AST: Audio Spectrogram Transformer (Gong et al., MIT, 2021)	First convolution-free audio model: treat the log-mel spectrogram as an image and apply a ViT.	The audio → vector path and the concrete preprocessing (16 kHz → 128-bin log-mel → 204 frames → ViT encoder).
4	Multimodal Diffusion for Text–Image–Audio (Wang, Zhang & Jiang)	Tri-modal diffusion with a hierarchical cross-modal alignment network and a tri-modal semantic alignment score.	The TSAS idea itself, and the framing that coherence must be measured between the three outputs, not just against the prompt.
5	Enhance Multimodal Consistency and Coherence (Lu et al., Penn State, ACL Findings 2025)	Scores text–image pairs on two channels — embedding metric and LLM judge on a 1–5 rubric — validated against humans with Cohen’s κ .	The two-channel evaluation (LLM judge + learned weights) and the calibration habit of hand-scoring a small batch.
6	Everything Can Be Described in Words / UMaT (Bi & Xu, arXiv 2503.09081)	Convert image and audio into unified text, then align and reason in text space; adaptive filtering removes redundancy.	The text-mediated coherence check, and the image↔audio text bridge in multimedia_score.py.
7	Cross-Modal Redundancy and the Geometry of V–L Embeddings (Dhimoila et al., ICLR 2026)	Iso-Energy Assumption: unimodal “atoms” act as modality-specific bias and fully explain the modality gap.	The explanation for our $s1 \gg s2/s3$ finding, and the motivation for the mismatched-pair baseline and per-pair floor.
8	ToMCLIP: Topological Alignment of Shared V–L Space (You et al., AISTATS 2026)	Use persistent-homology constraints to align the global geometry of the shared space, not just instances.	Conceptual support that pairwise cosine misses global structure — a pointer for geometry-aware future work.
9	Good Scores, Bad Data / MCS (independent research)	Decompose coherence into named dimensions; learn dimension weights to maximise correlation with performance; no human labels.	Decomposing coherence into named parts, learning weights by maximising correlation (our learn_weights and λ sweep), and localising the failure.
10	Likelihood-Ratio Biometric Score Fusion (Nandakumar et al., IEEE TPAMI 2008)	Optimal score-level fusion via the likelihood-ratio test; model genuine vs impostor score distributions.	The Likelihood-Ratio scorer and the Naive Bayes posterior — the principled replacement for the hand-tuned formula.

#	Paper	What it does	What we took
11	SGANet: Semantic & Geometric Alignment (Bai et al., HKUST-GZ)	Combines semantic and geometric alignment to flag cross-modal inconsistency as an anomaly.	Framing incoherence as anomaly detection — the mental model behind the mismatched-pair baseline and the variance term.

11. Issues Found and Fixed — Running Log

Issue	Status	Notes
Image scorer crashed on every image (old transformers API)	Fixed	outputs now non-zero; works on both library versions
secrets.py shadowed Python's built-in module	Fixed	file renamed
MeloTTS returns HTTP 500 ~25% of calls	Fixed	3-try retry + backoff, also in the live app
Image endpoint returns JPEG bytes in .png files	Known / harmless	libraries sniff the real format
Audio is 44.1 kHz, not 48 kHz	Handled	scorers resample per-scorer
Original WER compared transcript to the short prompt	Fixed	compare against the spoken paragraph → WER ≈ 0.095
Text output files carry a header the scorers read	Open	feeds the header into the WER term → 1-WER collapses to ≈0.19; strip header and re-run Phases 2–4
text_scores.csv is all zeros	Open	stale run; real Q_text ≈ 0.848 comes from quality_scores.csv
Learned weights barely generalise (LOO R ² ≈ 0.04)	Open — needs data	53 samples, 3 clustered features; treat weights as indicative

12. What Is Still Missing and Next Steps

12.1 Methodology

- **Reproducibility.** Generation sets no temperature, so every model runs at its default (~0.7) and the same prompt gives different output — and different scores — each run. Setting temperature = 0 for the benchmark runs would make text generation reproducible.
- **Repetition and variance.** Every prompt is scored from a single generation. Running each prompt three to five times and reporting mean ± standard deviation would make the numbers far more trustworthy.
- **Data-driven calibration.** The thresholds are borrowed rather than derived from our own labelled data; a calibration step would put them on firmer ground.
- **Human validation.** The coherence score has never been checked against human judgement. Hand-scoring 20–30 of the 53 triples for “do these three match,” then correlating each candidate method against that, is what will actually decide which method we propose.

12.2 Concrete To-Do Before the Next Review

- Strip the header from the text output files and re-run Phases 2–4 so WER, Q_audio, and the ImageBind similarities are computed on the real generated text.
- Re-run text_scores.csv so it is no longer stale.
- Finish the Naive Bayes and Likelihood-Ratio scorers end to end and add the mismatched-pair baseline as a standard column.
- Build the small hand-labelled coherence set and rank every candidate method against it and against the LLM-judge score.
- Take the winner — WTSAS on current evidence — to the professor for approval.

None of these gaps undermine the work so far. They are the natural next steps, and naming them plainly is part of doing the evaluation honestly.